Lavagne di Analisi III

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Ing. Fisica

Politecnico di Milano

I. FRAGALÀ

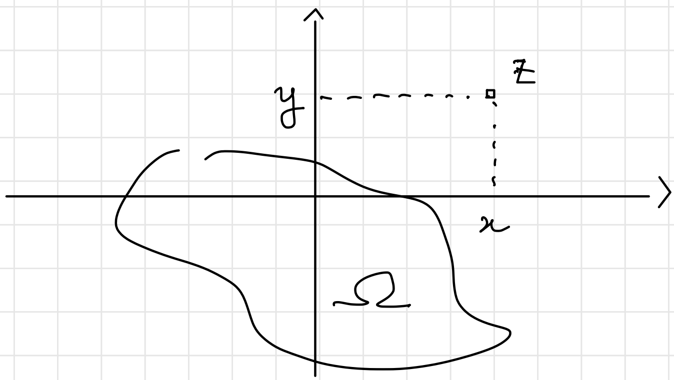
ANALISI COMPLESSA

<u>Def.</u> Una funzione di variabile complessa è una funzione $f : \Omega \subseteq \mathbb{C} \longrightarrow \mathbb{C}$

$$z = x + iy \qquad x, y \in \mathbb{R} \qquad i^2 = -1$$

$$f(z) = u(z) + i\,v(z) = u(x,y) + i\,v(x,y)$$

$$f \longleftrightarrow (u, v) : \Omega \subseteq \mathbb{R}^2 \longrightarrow \mathbb{R}^2$$



Esempi di funzioni elementari

- $f(z) = z^0 = x^0 + iy^0 \qquad \forall z \in \Omega = \mathbb{C}$

  $u(x,y) = x^0 \quad , \quad v(x,y) = y^0$

- $f(z) = \underset{x+iy}{\overset{\shortparallel}{z}} \qquad \forall z \in \mathbb{C}$

  $u(x,y) = x \quad , \quad v(x,y) = y$

- $f(z) = Re(z) \qquad \forall z \in \mathbb{C} \qquad f(x+iy) = x$

  $u(x,y) = x \quad , \quad v(x,y) = 0$

- $f(z) = Im(z) \qquad \forall z \in \mathbb{C} \qquad f(x+iy) = y$

  $u(x,y) = y \quad , \quad v(x,y) = 0$

- $f(z) = |z| \qquad \forall z \in \mathbb{C} \qquad f(x+iy) = \sqrt{x^2+y^2}$

  $u(x,y) = \sqrt{x^2+y^2} \quad , \quad v(x,y) = 0$

- $f(z) = P(z) = \underset{\uparrow}{a_n} z^n + \underset{\uparrow}{a_{n-1}} z^{n-1} + \dots + \underset{\uparrow}{a_1} z + \underset{\uparrow}{a_0}$

coefficienti $\in \mathbb{C}$ $\qquad \forall z \in \mathbb{C}$

$$f(z) = z^2 + iz + 1$$

$$= (x+iy)^2 + i(x+iy) + 1$$

$$= x^2 - y^2 + 2ixy + ix - y + 1$$

$$= (\underbrace{x^2 - y^2 - y + 1}_{u(x,y)}) + i(\underbrace{2xy + x}_{v(x,y)})$$

- $f(z) = \dfrac{P(z)}{Q(z)}$ $\qquad P, Q$ funzioni polinomiali

$\forall z \in \Omega = \{ z \in \mathbb{C} : Q(z) \neq 0 \}$

$$f(z) = \frac{z^2 + iz + 1}{z^2 - 1} \qquad \forall z \in \Omega = \{ z : z \neq \pm 1 \}$$

$$f(z) = \frac{z^2 + iz + 1}{z^2 + 1} \qquad \forall z \in \Omega = \{ z : z \neq \pm i \}$$

- $f(z) = e^z, \ \sin z, \ \cos z, \ \sinh z, \ \cosh z$

Funzione $f(z) = e^z$

$$Z = x + iy \implies e^z = e^{x+iy} := e^x \cdot e^{iy} = e^x(\cos y + i \sin y)$$

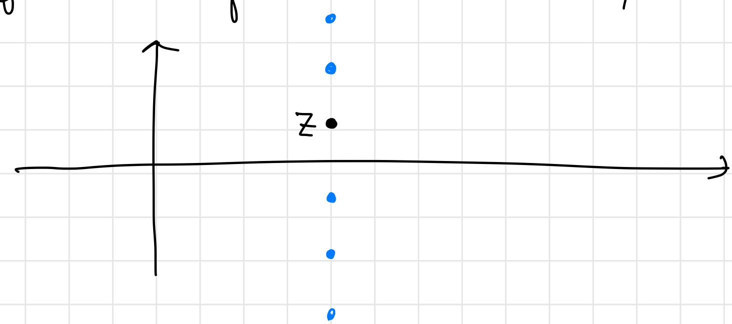
- $y = 0 \implies e^z = e^x = e^{\operatorname{Re} z}$
- $e^{z_1+z_2} = e^{z_1} \cdot e^{z_2}$

$$z_1 = x_1 + iy_1$$
$$z_2 = x_2 + iy_2$$
$$(z_1 + z_2) = (x_1 + x_2) + i\,(y_1 + y_2)$$

$$e^{z_1} \cdot e^{z_2} = e^{x_1}(\cos y_1 + i \sin y_1)\, e^{x_2}(\cos y_2 + i \sin y_2)$$
$$= e^{x_1+x_2}\left(\cos(y_1+y_2) + i \sin(y_1+y_2)\right).$$

- $u(x,y) = e^x \cos y, \qquad v(x,y) = e^x \sin y$
- $e^{z + 2\pi i} = e^{x + i(y + 2\pi)} = e^z$

$\Rightarrow$ la funzione exp è PERIODICA di periodo $T = 2\pi i$



- $e^z = 0 \iff |e^z| = 0$

$$|e^z| = |e^x|\,|\cos y + i \sin y| = e^x = e^{\operatorname{Re} z} \neq 0 \quad \forall z \in \mathbb{C}$$

Funzioni circolari

$$\cos z := \frac{1}{2}\left(e^{iz} + e^{-iz}\right), \quad \sin z := \frac{1}{2i}\left(e^{iz} - e^{-iz}\right)$$

- $y = 0$, $z = x \in \mathbb{R}$

$$\cos z = \frac{1}{2}\left(e^{ix} + e^{-ix}\right) = \frac{1}{2}\left(\cos x + \cancel{i \sin x} + \cos x \cancel{- i \sin x}\right)$$
$$= \cos x$$

- $\cos(z + 2\pi) = \cos(z)$ periodica di periodo 2π

- $$\cos(x+iy) = \frac{1}{2}\left(e^{i(x+iy)} + e^{-i(x+iy)}\right)$$
$$= \frac{1}{2}\left(e^{ix-y} + e^{-ix+y}\right)$$
$$= \frac{1}{2}\left[e^{-y}(\cos x + i \sin x) + e^{y}(\cos x - i \sin x)\right]$$
$$= \cos x \underbrace{\left(\frac{e^{y}+e^{-y}}{2}\right)}_{\cosh y} - i \sin x \underbrace{\left(\frac{e^{y}-e^{-y}}{2}\right)}_{\sinh y}$$

$$\Rightarrow u(x,y) = \cos x \cosh y, \quad v(x,y) = -\sin x \sinh y$$

- $\boxed{\cos z = 0}$

$$\begin{cases} \cos x \cosh y = 0 \\ \sin x \sinh y = 0 \end{cases}$$

1^A eq. $\Rightarrow \cos x = 0 \Rightarrow$
2^A eq. $\Rightarrow \sinh y = 0 \Rightarrow$

$$\boxed{\begin{array}{l} x = \frac{\pi}{2} + k\pi \qquad k \in \mathbb{Z} \\ y = 0 \end{array}}$$

- $\cos(iy) = \frac{1}{2}\left(e^{i(iy)} + e^{-i(iy)}\right)$

$$= \frac{1}{2}\left(e^{-y} + e^{y}\right) = \cosh y$$

$$\boxed{\cos(iy) = \cosh(y)}$$

- $\boxed{\cos^2 z + \sin^2 z = 1 \qquad \forall z \in \mathbb{C}}$

$\sin z$ propietà analoghe

$$\begin{cases} u(x,y) = \sin x \cdot \cosh y \\ v(x,y) = \cos x \sinh y \end{cases}$$

Funzioni iperboliche

$$\cosh z := \frac{e^{z} + e^{-z}}{2} \qquad \sinh z := \frac{e^{z} - e^{-z}}{2}$$

$$\begin{cases} \cosh z = \underbrace{\cosh x \cos y}_{u} + i \underbrace{\sinh x \sin y}_{v} \\ \sinh z = \underbrace{\sinh x \cos y}_{u} + i \underbrace{\cosh x \sin y}_{v} \end{cases}$$

periodiche di periodo $2\pi i$

⚠ Funzioni inverse ⚠